Visual Analytics Project Proposal

Leonardo Idone, 2088501

In this proposal we have chosen to create a visual analytics tool to visualize and explore F1 races in a data-driven fashion. To do so, we will leverage the publicly available data output by the FIA during races, aggregated by various libraries that archive them for historical purposes. Specifically, we have chosen the Python library Fast-F1 for its richness of data and high polling rate (sometimes as low as 3 Hz). The data has to be downloaded and extracted to a CSV file before use. Each separate session in an event is visualized individually. As an example of the expected data size, consider the "Emilia Romagna Grand Prix 2025"; its AS for all represented data stands at around: $AS = 662461 \times 11 = 7\,287\,071$. The program will be able to visualize every race from 2021 onwards (when the sensor data became publicly available), but for demonstration purposes we will use the aforementioned GP due to limited secondary storage space. If the complete set of data would be excessive, forms of reduction (e.g. increasing the coarseness of the timestamp by aggregating for deciseconds or centiseconds, diminishing precision of some columns) could be employed to gain performance on computations. All the visualizations will be synchronized with a central timeline that the user can seek, updating the displayed analytics accordingly. The software will display the track of interest, with drivers indicated as small dots moving along it according to the current timeline position. Together with it, the current track state and weather information will be shown (e.g., safety car status). Beside this, there will be a list of the top 10 drivers (those who scored points). By selecting either the dots on the track or the driver names, the user can filter the visualizations to only those selected drivers. Some visualizations will become accessible only after such a selection. For split times, a heat map will show the positive or negative time delta for each track sector and the filtered drivers. The user can click on any square to seek the timeline to the start of the corresponding sector. The deltas aren't available by default, but need to be calculated from the absolute lap time/absolute sector time depending if we are dealing with lap or sector times. Four coordinated line plots and a Gantt-like chart will display the following variables over time: throttle position, engine RPM, brake/DRS engagement (mutually exclusive), gear shifts, and speed. These are presented in separate plots to allow users to compare multiple drivers without visual clutter for each variable. Clicking on any part of the diagrams will move the timeline to the selected instant. Being as the track status, pit strategy, driver endurance and car state changes lap by lap usually, an interesting form of PCA would be for each lap seeked by the user, to aggregate all drivers state variables. Then, see for each driver what kind of different "styles" each pilot has chosen to adapt to the session situation. The way we aggregate the state variables (RPM, speed, nGear, Throttle, Brake, DRS, TyreLife) will be decided fully only after further research, as some variables would need to be converted to a continuous domain (Brake and DRS are boolean, for example), while considering the average of things like RPM, Speed, Throttle and nGear could prove acceptable. Again, to see how much each driver changes his driving style, as a form of adaptability, a different, global, PCA could be implemented in which we aggregate all results for each lap: drivers which prefer to remain consistent even under diverse situations (rain, old tyres, car damage) will be around the same area for each of their laps. Adaptable drivers will have many, different smaller clusters for each "mode" they decide to follow given the current situation. For the non-global view, a recomputation would be executed each time the user brushes to a new lap. Since clustering is a main goal, t-SNE might be a preferrable technique for our purposes. This will be decided after further testing. Finally, another chart compacts information about the tyres: the tyre compound used, its age at the time of a pit stop, the stint number (the sequence of the tyre set in the session), the pit-in time, and the pit-out time. Clicking on any element will move the timeline to that timestamp. The intended users of this software can be diverse: professional racing teams who need to discuss, present, or analyze a race in a clearer manner, journalists wanting to highlight interesting aspects of a race, or simply fans who enjoy exploring data from their favourite races. Note: all elements shown in the draft are preliminary and may contain inaccuracies; the representation only illustrates one possible UI layout for the aforementioned visualizations. Legends, labels, and details may be missing and should not be considered representative of the final project.

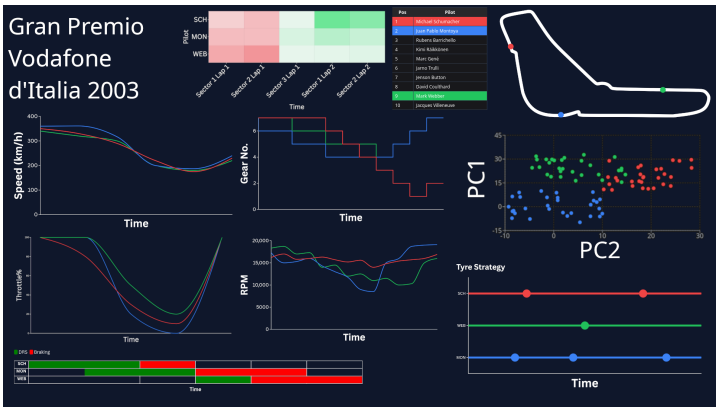


Figure 1: Mockup of the user interface.